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CSET203: Microprocessors and Computer Architecture (Even Sem 2024-25)

CSET-203_Mid-Semester Exam

Started on Wednesday, 12 March 2025, 10:19 AM

State Finished

Completed on Wednesday, 12 March 2025, 10:35 AM

Time taken 15 mins 36 secs

Grade 28.00 out of 30.00 (93.33%)

Question 1

Correct

Mark 1.00 out of 1.00

Which statement about the stack in 8085 is true?

- ☐ a. The stack pointer always points to the first empty location
- ☒ b. The stack grows towards lower memory addresses ✓
- ☐ c. The stack is a FIFO (First In, First Out) structure
- ☐ d. The stack grows towards higher memory addresses

The correct answer is: The stack grows towards lower memory addresses



Question 2

Correct

Mark 1.00 out of 1.00

The 8085 microprocessor has how many pins?

- ☐ a. 20
- ☐ b. 64
- ☒ c. 40 ✓
- ☐ d. 16

The correct answer is: 40

Question 3

Correct

Mark 1.00 out of 1.00

Which instruction copies the contents of the accumulator to the memory location specified by the contents of the register pair?

- ☐ a. STA
- ☐ b. LDA
- ☐ c. MOV
- ☒ d. STAX ✓

The correct answer is: STAX

Question 4

Incorrect

Mark 0.00 out of 1.00

The clock frequency of the 8085 microprocessor is:

- ☒ a. 5 MHz ✗
- ☐ b. 2 MHz
- ☐ c. 3 MHz
- ☐ d. 10 MHz

The correct answer is: 3 MHz



Question 5

Correct

Mark 1.00 out of 1.00

Which of the following flag is not accessible by the programmer in 8085?

- ☐ a. Carry Flag
- ☐ b. Sign Flag
- ☒ c. Auxiliary Carry Flag ✓
- ☐ d. Zero Flag

The correct answer is: Auxiliary Carry Flag

Question 6

Correct

Mark 1.00 out of 1.00

Which of the following is a Control Instruction in 8085?

- ☐ a. MOV
- ☒ b. NOP ✓
- ☐ c. JMP
- ☐ d. ANA

The correct answer is: NOP

Question 7

Correct

Mark 1.00 out of 1.00

The RIM instruction in the 8085 is used for:

- ☒ a. Reading the status of interrupts ✓
- ☐ b. Storing data in registers
- ☐ c. Resetting the microprocessor
- ☐ d. Setting interrupt masks

The correct answer is: Reading the status of interrupts



Question 8

Correct

Mark 1.00 out of 1.00

What is the function of the SIM instruction in serial communication?

- ☐ a. To check for pending interrupts
- ☐ b. To read the serial input data
- ☒ c. To set the serial output data ✓
- ☐ d. To disable serial communication

The correct answer is: To set the serial output data

Question 9

Correct

Mark 1.00 out of 1.00

The memory address in a microprocessor system is specified using the:

- ☐ a. Accumulator
- ☒ b. Address Bus ✓
- ☐ c. Data Bus
- ☐ d. Control Bus

The correct answer is: Address Bus

Question 10

Incorrect

Mark 0.00 out of 1.00

Which of the following is a vectored interrupt in 8085?

- ☒ a. RST 5.5 ✗
- ☐ b. INTR
- ☐ c. All options are correct
- ☐ d. TRAP

The correct answer is: All options are correct



Question 11

Correct

Mark 1.00 out of 1.00

The instruction cycle of a microprocessor consists of which three main steps?

- ☐ a. Load, Store, Fetch
- ☐ b. Decode, Fetch, Execute
- ☒ c. Fetch, Decode, Execute ✓
- ☐ d. Read, Write, Execute

The correct answer is: Fetch, Decode, Execute

Question 12

Correct

Mark 1.00 out of 1.00

The instruction "ADC B" performs which of the following operations?

- ☐ a. Subtracts the contents of register B from the accumulator
- ☐ b. Adds the contents of register B to the accumulator
- ☐ c. Complements the contents of the accumulator
- ☒ d. Adds the contents of register B and Carry flag to the accumulator ✓

The correct answer is: Adds the contents of register B and Carry flag to the accumulator

Question 13

Correct

Mark 1.00 out of 1.00

Which instruction is used to read the status of interrupts 7.5, 6.5, 5.5 and read serial data input bit?

- ☒ a. RIM ✓
- ☐ b. SIM
- ☐ c. NOP
- ☐ d. HLT

The correct answer is: RIM



Question 14

Correct

Mark 1.00 out of 1.00

Which flag indicates an overflow condition from the most significant bit?

- ☐ a. Sign Flag
- ☐ b. Zero Flag
- ☐ c. Parity Flag
- ☒ d. Carry Flag ✓

The correct answer is: Carry Flag

Question 15

Correct

Mark 1.00 out of 1.00

The SIM instruction is used for:

- ☐ a. Checking the status of interrupts
- ☐ b. Resetting the microprocessor
- ☒ c. Enabling/Disabling maskable interrupts ✓
- ☐ d. Controlling memory access

The correct answer is: Enabling/Disabling maskable interrupts

Question 16

Correct

Mark 1.00 out of 1.00

What is the difference between CMP R and CPI #?

- ☒ a. CMP compares with a register, CPI compares with an immediate value ✓
- ☐ b. CMP compares with an immediate value, CPI compares with a register
- ☐ c. CMP is for signed numbers, CPI is for unsigned numbers
- ☐ d. CMP performs subtraction, CPI performs addition

The correct answer is: CMP compares with a register, CPI compares with an immediate value



Question 17

Correct

Mark 1.00 out of 1.00

What is the value of the Sign Flag if the MSB of the result is 0?

- ☐ a. 1
- ☐ b. Same as Zero Flag
- ☒ c. 0 ✓
- ☐ d. Undefined

The correct answer is: 0

Question 18

Correct

Mark 1.00 out of 1.00

What is an unconditional branch instruction?

- ☐ a. It stops execution
- ☒ b. It jumps to a new memory location regardless of any conditions ✓
- ☐ c. It executes logical operations
- ☐ d. It jumps only if a condition is met

The correct answer is: It jumps to a new memory location regardless of any conditions

Question 19

Correct

Mark 1.00 out of 1.00

The instruction "HLT" is used to?

- ☒ a. Halt the microprocessor ✓
- ☐ b. Reset all flags
- ☐ c. Perform no operation
- ☐ d. Enable interrupts

The correct answer is: Halt the microprocessor



Question 20

Correct

Mark 1.00 out of 1.00

Which flag is used to check the number of 1s in the result?

- ☐ a. Carry Flag
- ☐ b. Zero Flag
- ☒ c. Parity Flag ✓
- ☐ d. Sign Flag

The correct answer is: Parity Flag

Question 21

Correct

Mark 1.00 out of 1.00

The instruction "JZ 2034H" is executed when?

- ☐ a. Carry flag is set
- ☒ b. Zero flag is set ✓
- ☐ c. Parity flag is set
- ☐ d. Sign flag is set

The correct answer is: Zero flag is set

Question 22

Correct

Mark 1.00 out of 1.00

The instruction "EI" in 8085 is used for?

- ☐ a. Halting the processor
- ☒ b. Enabling interrupts ✓
- ☐ c. Disabling interrupts
- ☐ d. Performing no operation

The correct answer is: Enabling interrupts



Question 23

Correct

Mark 1.00 out of 1.00

What happens when the PUSH B instruction is executed?

- ☒ a. The contents of the B register are stored in memory, and SP is decremented ✓
- ☐ b. The stack pointer is incremented
- ☐ c. The contents of the accumulator are stored in memory
- ☐ d. The B register value is moved to the accumulator

The correct answer is: The contents of the B register are stored in memory, and SP is decremented

Question 24

Correct

Mark 1.00 out of 1.00

What does the HLT instruction do in 8085?

- ☒ a. Stops executing the program until an interrupt occurs ✓
- ☐ b. Clears all registers
- ☐ c. Moves data from one register to another
- ☐ d. Halts execution and powers off the microprocessor

The correct answer is: Stops executing the program until an interrupt occurs

Question 25

Correct

Mark 1.00 out of 1.00

Which instruction is used to load the accumulator with data from a specified 16-bit address?

- ☒ a. LDA ✓
- ☐ b. MOV
- ☐ c. MVI
- ☐ d. STA

The correct answer is: LDA



Question 26

Correct

Mark 1.00 out of 1.00

What happens when the 8085 detects an interrupt?

- ☒ a. It completes the current instruction before handling the interrupt ✓
- ☐ b. It stops execution and resets the microprocessor
- ☐ c. It immediately jumps to the interrupt service routine
- ☐ d. It ignores the interrupt if another instruction is being executed

The correct answer is: It completes the current instruction before handling the interrupt

Question 27

Correct

Mark 1.00 out of 1.00

How many bytes are required for an instruction that contains an immediate 16-bit address?

- ☐ a. 1 byte
- ☐ b. 4 bytes
- ☐ c. 2 bytes
- ☒ d. 3 bytes ✓

The correct answer is: 3 bytes

Question 28

Correct

Mark 1.00 out of 1.00

Which instruction is used to perform logical AND operation with the accumulator and an 8-bit data?

- ☐ a. XRI
- ☒ b. ANI ✓
- ☐ c. ORI
- ☐ d. CPI

The correct answer is: ANI



Question 29

Correct

Mark 1.00 out of 1.00

The term "Micro" in microprocessor refers to:

- ☐ a. The use of miniature transistors
- ☐ b. The ability to process data at a very high speed
- ☒ c. The small size of the processor chip ✓
- ☐ d. A specialized function of the processor

The correct answer is: The small size of the processor chip

Question 30

Correct

Mark 1.00 out of 1.00

The instruction "MOV B, C" in 8085 represents?

- ☒ a. Copies the contents of register C into register B ✓
- ☐ b. Copies the contents of register B into memory
- ☐ c. Transfers data from memory to register B
- ☐ d. Moves the contents of register B into register C

The correct answer is: Copies the contents of register C into register B

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